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PAPER

Improved accuracy for Raman spectroscopic determination of polyethylene property by optimization of measurement temperature and elucidation of its origin by multiple perturbation two-dimensional correlation spectroscopy

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A novel strategy is demonstrated to improve the accuracy for determination of polyethylene (PE) density using Raman spectroscopy by optimizing the temperature of sample measurement. Spectral features associated with the conformation change of the polymer induced by temperature may provide valuable information to quantify important polymer properties such as density. To evaluate possible existence of an optimal temperature providing improved quantitative accuracy, Raman spectra of PE pellets with different densities were collected at eight different temperatures from 30 to 100 °C at 10 °C intervals. Using the spectral datasets collected at each temperature, partial least squares (PLS) models were developed using the reference PE density values determined by a standard density gradient method at 23 °C. Interestingly, the most accurate determination of density was realized at 70 °C. Multiple perturbation two-dimensional (MP2D) correlation analysis and differential scanning calorimetry (DSC) were used to examine the origin of improved accuracy at 70 °C. From these analyses, the pre-melt behavior of the PE samples was identified below their melting temperatures. Structural variations induced at the pre-melt stages enhance Raman spectral selectivity among the samples, thereby providing more accurate determination of PE density. The MP2D correlation analysis revealed the unforeseen thermal behavior of PE samples and successfully explained the improved accuracy at 70 °C.

Introduction

Previously, wide area illumination (WAI) Raman spectroscopy was demonstrated as a superior method to diffuse reflectance near-infrared (NIR) spectroscopy for measuring the density of polyethylene (PE) pellets.[1] Selective Raman spectral features were beneficial to describe the density variation of PE samples more precisely and the resulting accuracy was improved to the standard error of prediction (SEP) of 0.00083 g cm^{-3}. The above mentioned Raman measurement was performed by maintaining the samples at room temperature. Meanwhile, vibrational features of polymers are known to be highly sensitive to variations in temperature.[2–4] This structural sensitivity to temperature raises the interesting possibility of improving the accuracy of Raman measurement by collecting sample spectra at temperatures other than room temperature. Although the outcome cannot be predicted *a priori*, it is reasonable to assume that certain spectral features within the Raman spectrum collected at elevated temperatures would be more sensitive and/or selective to the property variation of polymers.

The objective of the work reported here is to explore an optimal temperature of PE pellets for Raman measurement that can improve the accuracy of density determination. Initially, the reference densities of PE pellets to correlate with the corresponding Raman feature were determined using the ASTM D1505-3,[5] based on the density gradient tube method. In this method, density was finally measured by controlling temperature of a sample at 23 °C in a constant-temperature bath. Then, Raman spectra of twenty five PE pellets of different grades were collected at eight different temperatures from 30 to 100 °C at 10 °C intervals. Changes in the intensity of the Raman bands, broadening of these bands, and shifting of peak positions simultaneously occurred as a function of sample temperature. Using the collected Raman spectra at each temperature, partial least squares (PLS) regression[6] was used to generate a multivariate calibration model to determine the densities of samples. Prediction errors for these models were compared to identify an optimal sample temperature providing improved PLS accuracy.

To possibly explain the spectral features responsible for the observed differences in density predictions as a function of

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temperature, a new type of two-dimensional (2D) correlation analysis[7–9] called multiple perturbation two-dimensional (MP2D) correlation analysis was used. This method was proposed by Shinzawa *et al.* to effectively study molecular behaviors in a complex chemical system.[10] The MP2D correlation analysis involves applying multiple and simultaneous perturbations as a means to induce unique structural variations in a system. Diverse molecular behaviors induced by simultaneous multiple perturbations can provide valuable clues to understand the characteristic structural variation in a system. The density as well as temperature changes were the multiple perturbations applied to the PE samples in this study. The mathematical description of MP2D correlation analysis can be found in a previous paper.[10]

To examine the variation of dissimilarity among the samples that could be induced by temperature elevation, we used multi-perturbation (MP) sample–sample 2D correlation analysis, further extended from conventional sample–sample 2D correlation analysis.[11,12] Particularly, MP sample–sample 2D correlation is well suited to extract intra-sample behavior such as thermal behavior based on multiple external perturbations. It efficiently provides information on the similarity or dissimilarity among samples. MP sample–sample 2D correlation highlights characteristic perturbation among samples, while MP2D correlation elucidates detail variations of spectral features.

Finally, differential scanning calorimetry (DSC) was also used to rationalize the results from the MP2D correlation analysis. The resulting thermograms were examined to identify possible conformational changes that coincide with findings from the MP2D correlation analysis. This DSC results illustrated how the pre-melting behavior of the polymer was related to accuracy of the density measurements using Raman spectroscopy.

Experimental

Samples and collection of Raman spectra

The twenty five PE samples used here were the same as those used and described before.[1] Briefly, the samples were composed of three different grades (homo PE, 1-butene copolymerized PE and 1-octene copolymerized PE) with the density range from 0.918 to 0.961 g cm^{-3}. Reference density of each sample was determined by using a standard test method as described in the ASTM D1505-3.[5] In this method, density reading of a sample was accomplished by maintaining temperature of a sample at 23 °C in a constant-temperature bath. As specified in the ASTM D1505-3, the within-laboratory standard deviation tested with 7 different PE samples ranged from 0.00018 to 0.00053 g cm^{-3} (average: 0.00034 g cm^{-3}). Average densities of homo-polymerized, 1-butene copolymerized and 1-octene copolymerized PEs were 0.958, 0.920 and 0.925 g cm^{-3}, respectively.

Raman spectra were collected with the wide area illumination (WAI) scheme (Kaiser Optical Inc., Ann Arbor, MI, USA) as used in our previous paper.[1] An excitation laser (785 nm, diode laser) was magnified to form a circular illumination area with a diameter of 6 mm (illumination area: 28.3 mm^2). The back-scattered Raman radiation was then collected by an array of 50 optical fibers and delivered to a CCD detector. PE pellets were transferred into a circular glass vial (diameter: 9.8 mm). Then, this vial was positioned in a metal block capable of controlling the temperature by feedback control to a precision of ±0.1 °C. The temperature of a sample was checked by directly inserting a thermocouple inside the PE packing. After confirmation of sample temperature, the thermocouple was removed for Raman spectral collection.

Raman spectra were collected for each sample at eight different temperatures from 30 to 100 °C, with an interval of 10 °C. Triplicate spectra were collected for each sample. Before collecting each spectrum, the sample vial was shaken vigorously to redistribute the pellets, thereby ensuring random pellet packing for each measurement. All chemometric algorithms were performed by in-house software prepared with MATLAB (Version 7.01; The Math Works, USA).

Differential scanning calorimetry (DSC)

DSC was also performed on the same set of PE samples over a temperature range from 30 to 150 °C. The measurements were performed with a TA Instruments model Q100 system (New Castle, DE, USA). For the DSC analysis, approximately 3 mg of sample taken from each pellet was transferred into a standard DSC aluminium pan for analysis. The initial temperature was set to 30 °C and the sample was heated at a rate of 10 °C min^{-1} until a final temperature of 150 °C was reached. An empty pan was used for reference. The resulting thermograms were recorded and analyzed subsequently.

Results and discussion

Spectral variations at different temperatures

Fig. 1(a) shows normalized Raman spectra (1504–1054 cm^{-1} range) of a PE sample (density: 0.918 g cm^{-3}) collected at the eight different temperatures. Baselines of raw Raman spectra were corrected at five different wavenumbers (1504, 1402, 1196, 1114 and 1054 cm^{-1}) and the peak area was calculated under the 1504–1054 cm^{-1} range.[1] Each baseline corrected spectrum was normalized by the corresponding peak area. Variations in the spectral bands as a function of temperature are highlighted in three narrower spectral ranges shown in Fig. 1(b)–(d), corresponding to the following spectral ranges, respectively, 1504–1350, 1330–1270 and 1190–1057 cm^{-1}. These spectra are offset for clarity and presented sequentially with spectra collected at 30 and 100 °C plotted on the top and bottom, respectively. Vibrational assignments for the major bands are summarized in Table 1. The observed band positions present in Table 1 are well-matched with those observed in previous publications,[13–16] in which Raman spectroscopy had been used extensively to study the polymeric structure of PE.

As shown Fig. 1(b)–(d), the intensities of major bands decrease and their bandwidths broaden with the increase of temperature. As the temperature increases, the molecular structure of the polymer becomes more disordered, thereby resulting in the observed broadening of the spectral features. The most significant temperature-induced variations in terms of band shapes occur in the 1500–1400 cm^{-1} region (see Fig. 1(b)) where the 1464, 1443 and 1420 cm^{-1} bands are highlighted. Intensities of the bands at 1464 and 1420 cm^{-1} become weaker at higher temperatures, while the intensity of the 1443 cm^{-1} peak is

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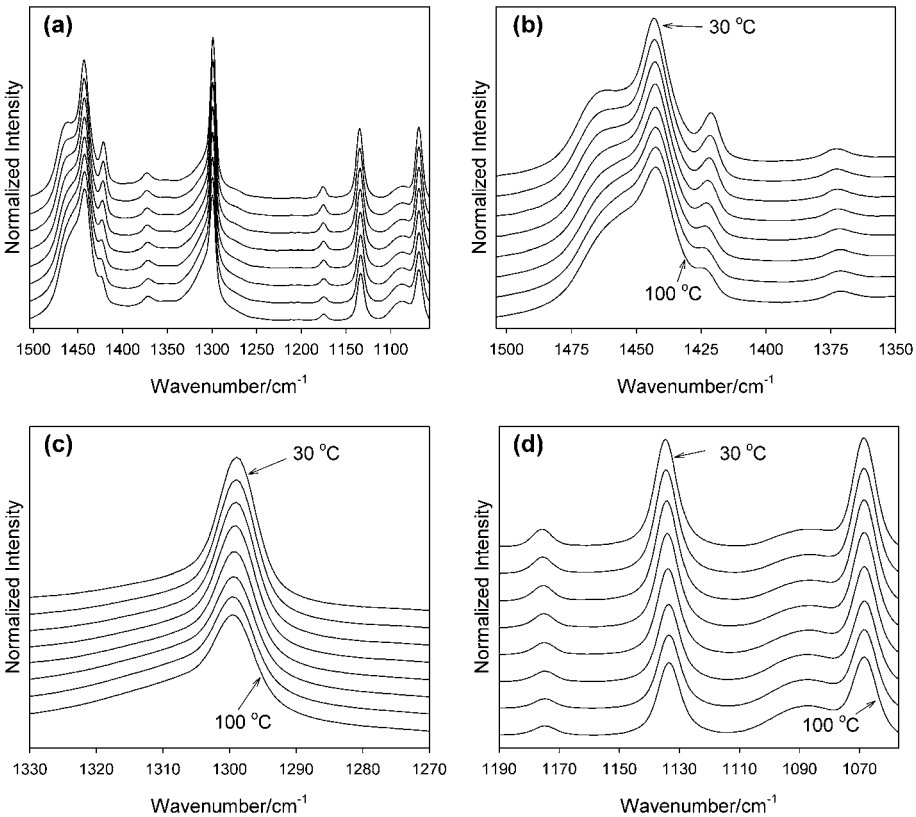

Fig. 1 Normalized Raman spectra of a PE polymer sample (density: 0.918 g cm^{-3}) collected at eight different temperatures (30–100 °C, 10 °C interval) in the 1504–1054 (a), 1504–1350 (b), 1330–1270 (c) and 1190–1057 cm^{-1} (d) spectral regions. Spectra are presented in sequence from 30 (top) to 100 °C (bottom). All the spectra in the figure are arbitrarily offset for clarity.

Table 1 Vibrational assignments of the observed Raman bands of PEs

Wavenumber/cm^{-1}	Vibrational mode
1464	CH_2 rocking
1443	CH_2 bending
1420	CH_2 bending
1372	CH_2 wagging
1298	CH_2 twisting
1138	C–C stretching
1070	C–C stretching

relatively unchanged. The band centered at 1298 cm^{-1} (CH_2 twisting) decreases in magnitude and simultaneously broadens with the increase of temperature as shown in Fig. 1(c). The same trends are also observed for the bands in the 1150–1057 cm^{-1} region, as shown in Fig. 1(d). In this region, the 1138 and 1070 cm^{-1} bands corresponding to the C–C (backbone) stretching modes are presented.

To evaluate the relationship between the degree of temperature-induced spectral variations and the density of a PE sample, we have selected five PE pellets with different densities (0.918, 0.928, 0.934, 0.955, 0.961 g cm^{-3}) and examined the corresponding Raman spectra collected at different temperatures, as shown in Fig. 2. In Fig. 2, only the spectra collected at 30 and 100 °C are shown for clarity. Normal and bold lines correspond to the spectra collected at 30 and 100 °C, respectively. In addition, difference spectra are also presented in Fig. 2 for the PE spectra collected at each temperature. These difference spectra were calculated by subtracting the spectrum collected at 30 °C from all other spectra.

As the temperature of the PE pellets is varied, intensity variations are most significant for the three bands centered at 1298, 1138 and 1070 cm^{-1} in the 1350–1054 cm^{-1} spectral window. Intensity variations in this window are relatively larger for the low density polymers as can be easily seen in the Raman spectra collected at two different temperatures of 30 and 100 °C. For the low density samples, polymeric structure is less ordered, so more freedom in molecular vibration is available inside a pellet and it eventually leads to larger spectral variation. In addition, the increased freedom in vibration leads to expand its volume. As known, the coefficient of thermal expansion for low density PEs is higher compared to that for high density PEs.[17]

Within the 1350–1250 cm^{-1} spectral region, the difference spectra illustrate that the magnitude of the difference decreases and the difference peak sharpens as the density of the polymer increases. This same trend is observed for the 1070 cm^{-1} band, while the difference pattern is somewhat different for the 1138 cm^{-1} band. As the density increases, the difference peak at the 1138 cm^{-1} band resembles a derivative feature, indicating a shift in the band position. For PE with the density of 0.918 g cm^{-3}, the difference peak is primarily negative for the 1138 cm^{-1} band, while both negative and positive features are evident for the PE with the density of 0.961 g cm^{-3}. This observation clearly indicates the occurrence of a band position shift. Overall, in the

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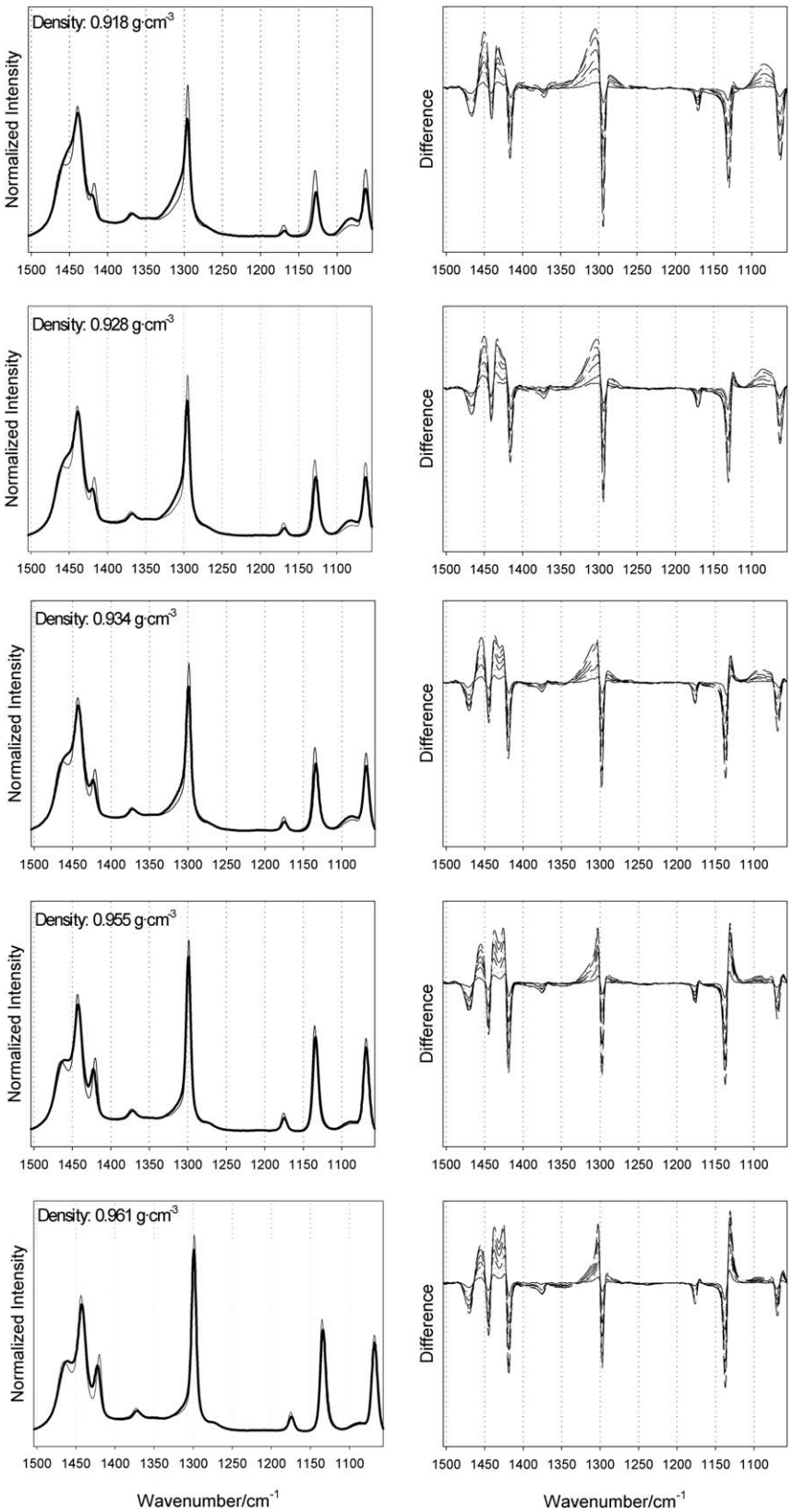

Fig. 2 Normalized Raman spectra of five PE polymers with different densities collected at 30 (normal line) and 100 °C (bold line). Difference spectra were generated by subtracting the spectra at 40–100 °C from the spectrum at 30 °C. Each difference spectrum is shown to the right of the corresponding Raman spectra.

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1350–1054 cm^{-1} spectral region, both the changes of band intensities and shifts of band positions synchronously occur as a function of polymer temperature.

Variations of three bands at 1464, 1443 and 1420 cm^{-1} in the 1504–1400 cm^{-1} region are quite different compared to those observed in the 1350–1054 cm^{-1} region. Band shape variations are more evident in the 1504–1400 cm^{-1} region. For the high density PE (0.961 g cm^{-3}), the three bands are still observable at 100 °C although the 1464 cm^{-1} band is fairly broadened, while for the low density PE (density: 0.918 g cm^{-3}) at 100 °C, the 1464 and 1420 cm^{-1} bands nearly disappear and only the 1443 cm^{-1} band is observable. The temperature-induced structural disorder is more significant for the low density polymers and leads to considerable broadening of the spectral features centered at 1464 and 1420 cm^{-1}.

Consequently, the temperature-induced Raman spectral variations of the PE pellets with the different densities clearly show different behaviors in terms of band intensities, band positions and band shapes. These spectral variations open the possibility that temperature could provide Raman spectra with more selectivity or/and higher sensitivity, thereby possibly improving accuracy of the density measurement.

Partial least squares to determine the density of PE pellets at different temperatures

Partial least squares (PLS)[6] regression was used to build multivariate calibration models to determine the density of PEs by using the Raman spectra collected at the eight different temperatures. As mentioned in the Experimental section, the reference density values determined at 23 °C using the standard ASTM method were used for PLS. As performed in the previous study,[1] the resulting errors were obtained by cross-validation with a set of 5 validation samples. Initially, 20 and 5 samples were assigned to the calibration and validation sets, respectively. There were 53 130 unique combinations of selecting 5 validation samples from the dataset of 25 samples. All possible combinations were tested and the mean standard error of calibration (MSEC) and the mean standard error of prediction (MSEP) were calculated by averaging the individual SEC and SEP values. The 1504–1054 cm^{-1} range was used for PLS and each model was generated with four factors. The results are presented in Table 2 for each PLS calibration model.

The MSEP varied as a function of pellet temperature, as shown in Fig. 3. A minimum MSEP was 0.00059 g cm^{-3} when the pellet temperature was maintained at 70 °C. Fig. 4 shows the prediction correlation plots obtained at room temperature (a) and 70 °C (b). Clearly, the correlation between reference density and predicted density is better when Raman spectra of the PE pellets are collected at 70 °C. The improved accuracy was statistically significant compared to that acquired at room temperature[1] based on a *t*-test at 95% confidence level ($P_{calculated} = 0.02 < 0.05 = P_{95\%}$). These findings indicate that the Raman spectra of PE pellets acquired at 70 °C are more informative to follow the variation in densities of the same samples at 23 °C.

It is difficult to rationalize the optimal temperature for density measurement by simple inspection of the spectra presented in Fig. 1 and 2. In order to probe this question more deeply and to identify spectral evidence for the optimum temperature at 70 °C, we initially examined the loadings for each PLS analysis. The first loading vector determined at each temperature is presented in Fig. 5 (a)–(c) for the 1470–1410, 1320–1280 and 1160–1054 cm^{-1} spectral ranges, respectively. In all three ranges, variations in the first loading vector show similar trends with temperature. As the sample temperature increases, the magnitude of the weighting vector decreases and the corresponding band shape broadens. In addition, the band positions shift either to a lower or a higher wavenumber in accordance with the temperature variation. Typically, the shape of the first loading vector resembles the average spectrum of a given dataset; therefore, variation of first loading is similar to variations in the spectral features presented in Fig. 1 and 2. Based on the examination of variations in the first loading vector, no specific clues to rationalize the optimal temperature are found.

Fig. 6 (a)–(c) show the corresponding second loading vectors determined for each temperature over the same three spectral regions. As shown in these figures, the second spectral loadings change dramatically as a function of temperature. Although the shapes of second loadings in the 1470–1410 cm^{-1} region (refer to Fig. 6(a)) vary continuously with increasing temperature, those from 30 to 50 °C (red, green, yellow colors) are relatively similar to each other. A second group of vectors similar in shape includes those at 60, 70, and 80 °C and a third grouping is evident for the vectors at 90 and 100 °C.

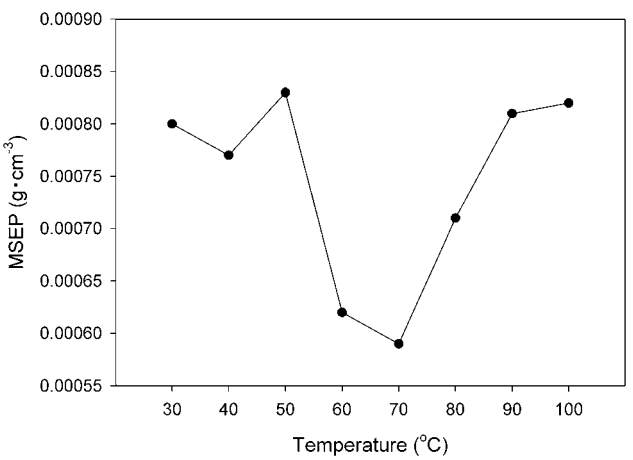

Fig. 3 Variation of MSEP acquired at eight different temperatures.

Table 2 Cross-validated PLS calibration results at eight different temperatures using the 1504–1054 cm^{-1} range with four factors in all the cases

	30 °C	40 °C	50 °C	60 °C	70 °C	80 °C	90 °C	100 °C
MSEC[a]	0.00058	0.00053	0.00060	0.00048	0.00048	0.00049	0.00045	0.00052
MSEP[a]	0.00080	0.00077	0.00083	0.00062	0.00059	0.00071	0.00081	0.00082

[a] Values presented in g cm^{-3}.

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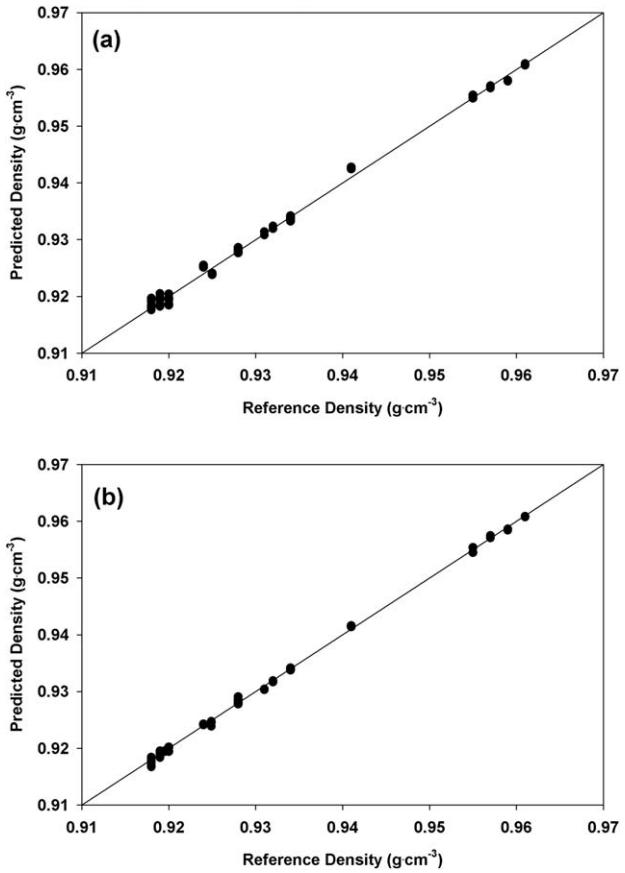

Fig. 4 Prediction correlation plots obtained at room temperature (a) and 70 °C (b).

Similar trends are evident over the 1320–1280 cm^{-1} spectral region (refer to Fig. 6(b)). Loading vectors from the eight different temperatures can be arranged into two groups in terms of the shape and position of the vector bands. The loadings from 30 to 50 °C are similar in shape, while the loadings from 60 to 100 °C are similar in shape but differ compared to those corresponding to the 30–50 °C vectors. The loading weight shifts to lower wavenumbers and becomes broader features for this high temperature group.

In the 1160–1054 cm^{-1} region (refer to Fig. 6(c)), abrupt changes in the loading vector shape are evident at 60 and 70 °C. Then, the loadings from 70 to 100 °C are similar. Inspection of the second loading vectors indicates noticeable variations by temperature occur abruptly around 60 or 70 °C.

The overall examination of the second loadings in three different spectral regions reveals that the temperature-induced structural variation seems to start around 60 or 70 °C with a subsequent variation after 90 °C. DSC analysis was performed to confirm the variations found by examination of the second loading vectors. The recorded DSC thermograms of all PE samples used in this study are presented in Fig. 7. The endothermic peaks corresponding to the characteristic melting temperature (T_m) of each sample appear over the temperature range from 105 to 140 °C. These thermograms are presented in three colors where black, blue and red colored thermograms correspond to lower (0.918–0.924 g cm^{-3}), middle (0.925–0.934 g cm^{-3}) and higher density (0.941–0.961 g cm^{-3}) PE pellets,

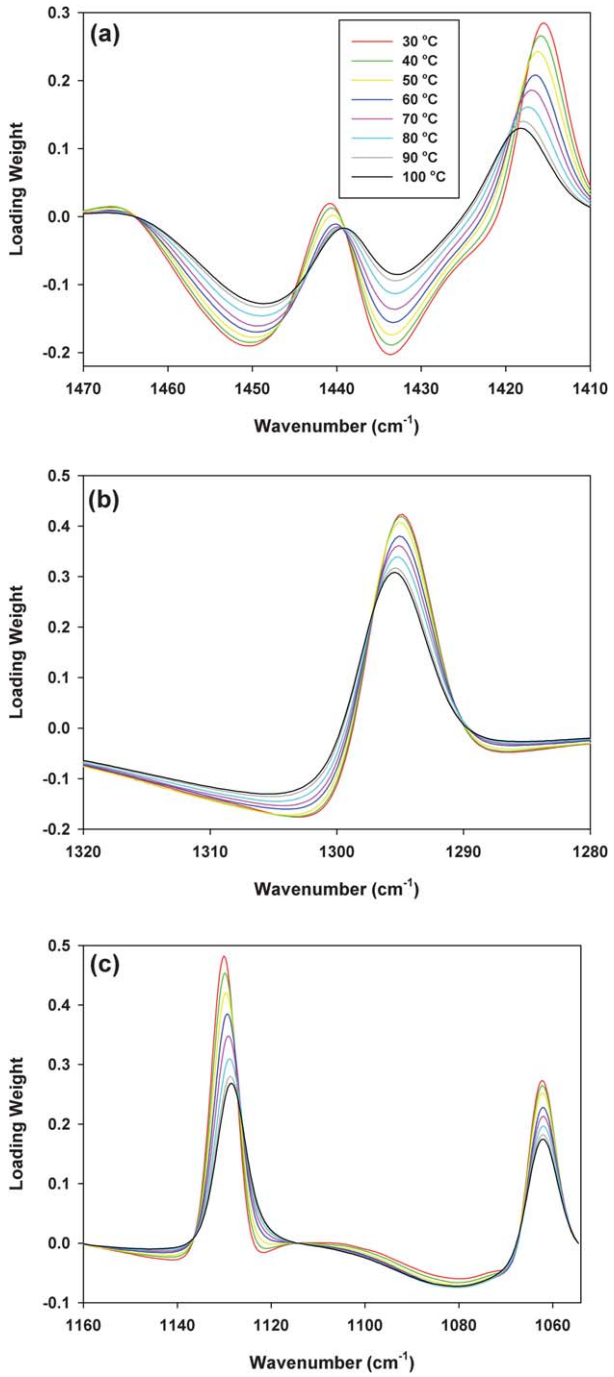

Fig. 5 The first PLS loading vectors at each temperature in the 1470–1410 (a), 1320–1280 (b) and 1160–1054 cm^{-1} (c) range. Eight different colors are used to differentiate each loading.

respectively. The lower density samples were mostly 1-butene or 1-octene co-polymerized PE samples. As expected, the lower density pellets have lower melting temperatures with broader peaks; while, the higher density PEs have higher melting temperatures with the narrower peaks. The inset inside Fig. 7 highlights the thermograms in the 40–100 °C range with the same color notation. As shown, small endothermic features are apparent for the lower density polymers (14 out of 25 total samples) from 60 to 90 °C range, while the phase changes

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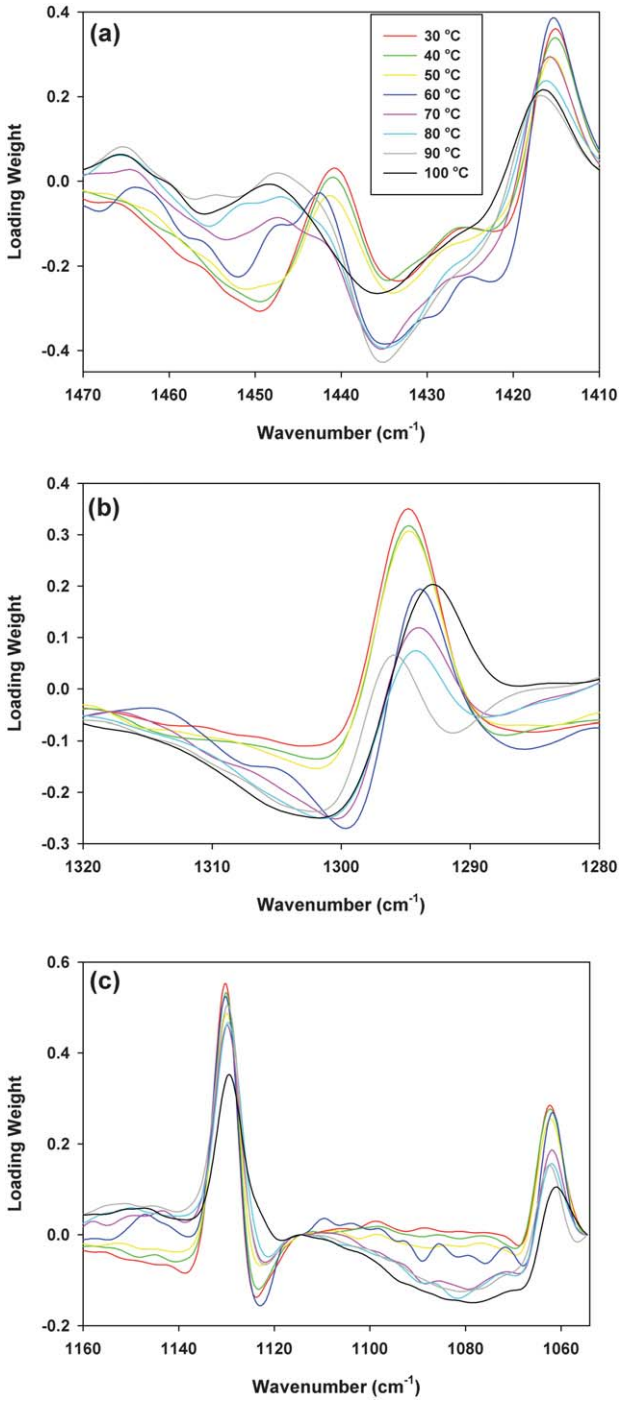

Fig. 6 The second PLS loading vectors at each temperature in the 1470–1410 (a), 1320–1280 (b) and 1160–1054 cm^{-1} (c) range. Eight different colors are used to differentiate each loading.

largely occur in between 70 and 80 °C. In contrast, no significant phase changes are observed for middle and higher density samples (11 out of 25 total samples). The DSC thermograms clearly demonstrate small phase changes in the lower density pellets even below their melting temperatures. DSC results are clearly supportive of the sudden change of the shapes in the second loading vectors at 60 or 70 °C, as shown in Fig. 6.

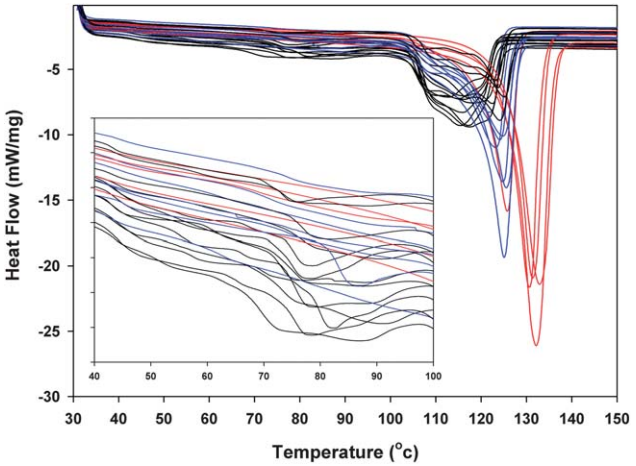

Fig. 7 DSC thermograms of all 25 PE samples used in this study. The temperature of samples was increased from 30 to 150 °C with a rate of 10 °C per minute. Black, blue and red colored thermograms correspond to lower, middle and higher density PE pellets, respectively. The inset highlights the thermograms in the 40–100 °C range with the same color notation.

Multiple perturbation 2D (MP2D) correlation analysis

MP2D correlation analysis[10] was individually performed over the three narrower spectral regions (1180–1100, 1320–1280 and 1504–1400 cm^{-1}). The spectra of the homo PEs of different densities collected at the eight different temperatures were used for the analysis. The synchronous and disrelation maps (not shown in this paper) obtained from 1180–1100 cm^{-1} and 1320–1280 cm^{-1} regions showed typical 2D correlation patterns. A well-known pattern indicating the shifting band that simultaneously loses intensity[9] was observed in the 1180–1100 cm^{-1} region. It implies a specific transformation of crystalline and amorphous structures. In the 1320–1280 cm^{-1} region, so-called a butterfly pattern in the disrelation map was observed. It indicates variations of two overlapping bands changing in intensity in opposite directions.[9]

The synchronous (a) and disrelation (b) maps in the 1504–1400 cm^{-1} region are shown in Fig. 8, where the CH_2 rocking and bending modes of the PE side chains are present. Although the interpretation of the patterns shown in the figure seems somewhat difficult due to the presence of broadening and shifting Raman peaks, the development of the complex cluster pattern in the 2D correlation maps suggests the presence of a conformational change of the crystalline structure of PE. The disrelation correlation intensity becomes larger, corresponding to more out-of-phase behavior for the side chains of PEs induced by the applied perturbations. The melting temperature (T_m) of PE-based polymers is higher than 100 °C, and therefore, the observed features in the correlation analysis are probably originated from a partial fusion of less ordered small crystalline domains.

Overall, synchronous and disrelation structural behaviors by both temperature and density perturbations are observed in the whole spectral range, while the most noticeable variations are observed in the 1504–1400 cm^{-1} region, representing complex variations of the PE side chains. Synchronous and disrelation

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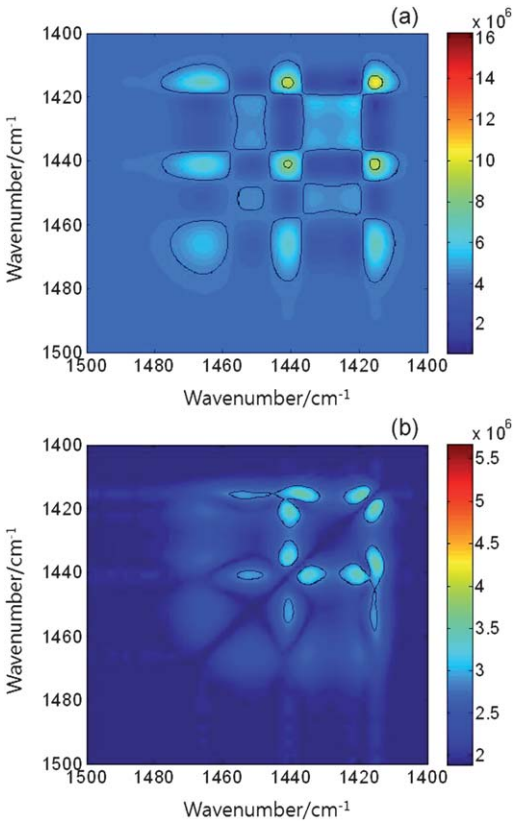

Fig. 8 Synchronous (a) and disrelation (b) correlation maps of homo PEs in the 1500–1400 cm^{-1} region.

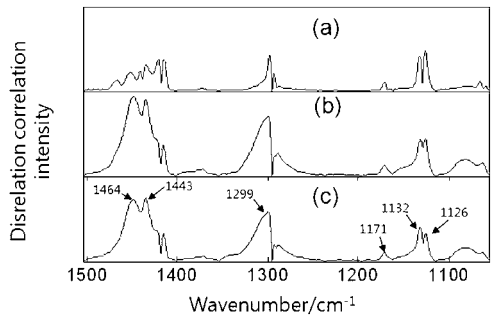

Fig. 9 Disrelation correlation spectra of homo (a), 1-butene copolymerized (b) and 1-octene copolymerized PEs (c) sliced at 1295 cm^{-1}.

correlation maps (not shown in this paper) generated from the sets of Raman spectra of 1-butene copolymerized PEs and 1-octene copolymerized PEs show similar correlation patterns compared to those of the homo PEs. Since all three grades of PEs undergo similar structural alterations, the more in-depth comparison of out-of-phase behavior among the samples could provide an important clue to understand the corresponding polymeric behavior based on the temperature perturbation.

Disrelation correlation spectra of the (a) homo, (b) 1-butene copolymerized and (c) 1-octene copolymerized PEs sliced at 1295 cm^{-1} (CH_2 twisting) are shown in Fig. 9. Each plot has the same *y*-scale for comparison of the disrelation correlation intensity. Disrelation correlation intensity is a measure of the degree of dissimilarity between changes in spectral intensity measured at ν_1 and ν_2. For example, the development of two peaks observed at 1132 and 1126 cm^{-1} (C–C stretching) indicates that the thermal behavior of the C–C backbone is different from that of the CH_2 twisting mode. Because disrelation correlation intensity is semi-quantitative in terms of the degree of dissimilarity, relatively similar levels of disrelation correlation intensities at 1132 and 1126 cm^{-1} among the three plots suggest the analogous thermal behavior of the C–C backbone structure among the three grades of samples.

The disrelation intensities and shapes in the 1500–1300 cm^{-1} region are quite different between the homo PE and the copolymer PEs, while these features are fairly similar for both copolymer PEs. Differences indicate that the thermal behavior of the CH_2 structures in the homo PE is different from that of the copolymer PEs. The CH_2 structures of two copolymers are more sensitively varying to the heating stimulus comparing to the homo PE even well below T_m. Semi-crystalline polymers such as LLDPE (liner low density polyethylene) are known to have complex supramolecular structures consisting of folded-chain crystal lamellae embedded in a liquid-like amorphous matrix,[18–21] which could orient to different directions under an external perturbation. Therefore, even well below the T_m, an LLDPE sample gradually undergoes a partial fusion of less ordered small crystallites upon heating, known as pre-melting.[20] Pre-melting is fundamentally similar to the normal melting process, except that it proceeds at temperatures lower than T_m. The presence of high disrelation correlations in the 1500–1300 cm^{-1} region is mostly due to the pre-melting process. These observations acquired from MP2D correlation analysis are well accordant with the results from DSC.

Multi-perturbation sample–sample 2D correlation analysis

Although MP2D correlation analysis successfully demonstrates pre-melting behavior lower than T_m, the findings do not fully explain the improved accuracy at 70 °C. To address this point, it would be worthwhile to examine the variation of dissimilarity among these samples. For this purpose, we used multi-perturbation (MP) sample–sample 2D correlation analysis, further extended from conventional sample–sample 2D correlation analysis.[10] The concept of synchronous sample–sample correlation involves spectral matching based on inner products of spectral vectors. Synchronous sample–sample correlation intensity becomes significant only if spectral features of samples are similar in shape. On the other hand, it becomes small when the spectral features of samples are substantially different. Particularly, MP sample–sample 2D correlation is well suited to extract intra-sample behavior such as thermal behavior based on multiple external perturbations. It efficiently provides information on the similarity or dissimilarity among samples.

The autopower spectra corresponding to the correlation intensity across the main diagonal position in the 2D synchronous correlation map are shown for the (a) homo, (b) 1-butene copolymerized and (c) 1-octene copolymerized PEs in Fig. 10, respectively. The variations of synchronous correlation intensities of three grades of PEs from 30 to 100 °C are similar to each other (generally U-shape). Correlation intensities are strong at both low and high temperatures, while it is lowest at 70 °C. The higher intensity indicates that the samples show relatively similar spectral variations by the temperature perturbation. The lowest

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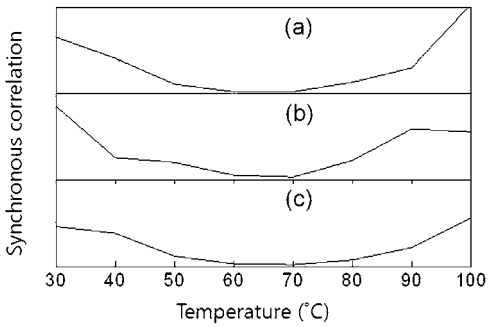

Fig. 10 The autopower spectra corresponding to the correlation intensity across the main diagonal position in the 2D synchronous correlation map for the (a) homo, (b) 1-butene copolymerized and (c) 1-octene copolymerized PEs.

intensity at 70 °C indicates that the samples are most structurally different from each other. This is due to the occurrence of pre-melting, corresponding to partial fusion of less ordered small crystalline domains.

The increase of dissimilarity at 70 °C among the samples suggests the presence of a specific transient state of the crystalline structure related to the conformational defect. The presence of co-monomers disrupts the packing of regular molecular chain structure and thus brings lower crystallinity. Watanabe *et al.* proposed a conformation model of PE that the side chains of co-monomers exist outside of a crystalline lattice and thus the release from structural constraint by such exclusion eventually leads to the formation of conformational defects at a relatively low temperature.[18] Importantly, the observation derived from the MP2D correlation analysis agrees well with this model.

The varied polymeric structure provides more spectroscopically discernible features among the samples. Therefore, the improved spectral selectivity among the samples provides a valuable clue to improve the performance of quantitative analysis. The overall results from the MP sample–sample 2D correlation analysis successfully describe the best PLS result for the determination of density using Raman spectroscopy when spectra of the samples are collected at 70 °C.

Conclusions

The best SEP achieved in this study by optimizing the temperature of Raman measurement at 70 °C was 0.00059 g cm^{-3}. Although it is still higher than the within-laboratory standard deviation (0.00034 g cm^{-3}) defined in the corresponding ASTM method, the resulting accuracy is still excellent compared to those from previous reports.[1,22–24] The proposed analytical scheme for density measurement of PE samples is simple since it only requires a temperature control device for Raman spectral collection. In addition, it is fast and non-destructive without demanding any sample preparation steps for analysis.

The demonstrated strategy of improving accuracy by maximizing sample dissimilarity through an application of a simple external perturbation could be applicable for quantitative analysis of other polymers. Along with the temperature perturbation, other perturbations that could effectively enhance sample dissimilarity would be also valuable.

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